V3.1

2000m

Miniature Laser Ranging Module User Manual

User Manual

Based on the TOF (Time of Flight) principle, this laser ranging module supports a ranging range of 2~2000m, and has UART (TTL_3.3V) communication interface, single ranging, continuous ranging and other functions.

Contents

1 Safety Warnings and Precautions

Warning

1.Direct eye view of the laser emitter is prohibited to avoid eye damage (Conform to the Class 1 standard).

2.Working environment temperature range: -20~ +70 ° C, avoid high temperature, humidity, rain, snow or dust environment.

3.It is forbidden to disassemble the module by yourself, otherwise it may void the warranty or damage the device.

4.This product is mainly suitable for the integration of civil UAVs, outdoor observation, engineering surveying and mapping and other equipment, and is prohibited from being used for various military equipment.

5.The spot adjustment mode is for developers only. Do not aim it at your eyes in this mode to avoid eye damage.

2 Product Overview

2.1 Product Introduction

Based on the TOF (Time of Flight) principle, this laser ranging module supports a ranging range of 2~2000m, and has UART (TTL_3.3V) communication interface, single ranging, continuous ranging and other functions,and complies with the EU RoHs requirements.

2.2 Technical Parameters

Basic Performance	Eye Safety	Class 1（IEC 60825-1）
	Wavelength	905±10nm
	Divergence	2.5×5mrad
	Receive FOV	~8mrad
	Max range (Visibility①≥5Km)	≥2000m @60% reflectivity, Building Target
		≥1400m @30% reflectivity, 2.3×2.3 m target
		≥1000m @30% reflectivity, 0.5×1.7 m target
		≥500m @30% reflectivity, 0.2×0.3 m target
	Min range	2m
	Ranging Accuracy	d≤800m:±(0.4m+ d×0.07%)m d>800m:±1m（d represents the distance, with the unit of m）

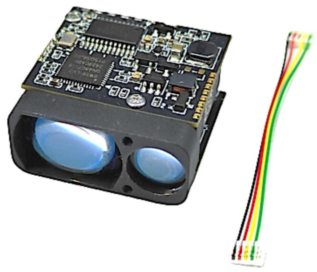
	Frequency	1~5Hz(Adaptive)
	Detection probability	⩾98%
	False alarm rate	⩽1%
Electrical Performance	Interface Type	UART(TTL_3.3V/2.5V)②
	Power Supply	DC 3~5V
	Standby Power	⩽0.5mW（Power on pull down)
		⩽0.3W（Power on pull up)
	Operating Power(+5Vin)	⩽0.70W@25°C
		⩽0.75W@70°C
		⩽0.6W@-20°C
	Startup Time	⩽200ms
Physical Performance	Weight	10.5±0.5g
	Dimensions	26×25×14mm（L×W×H)
	Shock Resistance	1200g，1ms
	Vibration Resistance	5~50~5 Hz,1 octave /min,2.5g
Environmental Adaptability	Operating Temp.	-20 ~ +70°C
	Storage Temp.	-25 ~ +75°C
	Reliability	MTBF⩾1500h

① Visibility refers to the visibility in hazy weather, and sandstorm weather has a greater impact on ranging performance.

② The actual signal is at a high level with a real voltage of 2.5V and is compatible with TTL 3.3V UART

3 Hardware Installation Guide

3.1 Packing List



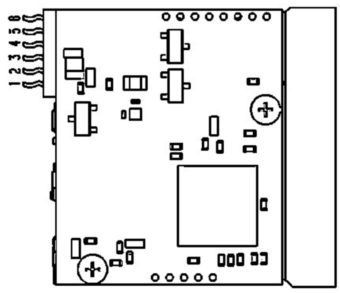
1.Laser Ranging Module×1

2.Power Cable×1

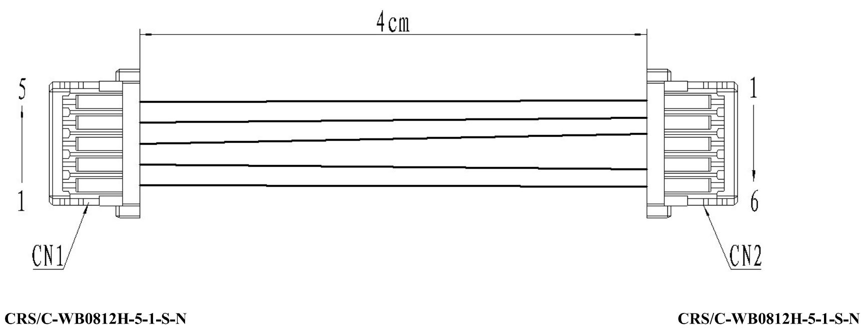
3.2 Interface Definitions

Interface Type:UART(TTL_3.3V)

Connector Model:FWF08002-S06B13W5M(TXGA Connector).

Pin 1	POWER_ON	Module Power Switch, TTL 3.3V Level; Module On (>2.7V), Module Off (<0.3V)	
Pin 2	UART_RX	Serial Port Receive Terminal, TTL 3.3V Level	
Pin 3	UART_TX	Serial Port Transmit Terminal, TTL 3.3V Level	
Pin 4	NC		
Pin 5	Power Positive	Power Supply: 3~5V	
Pin 6	GND	Serial Port Ground	

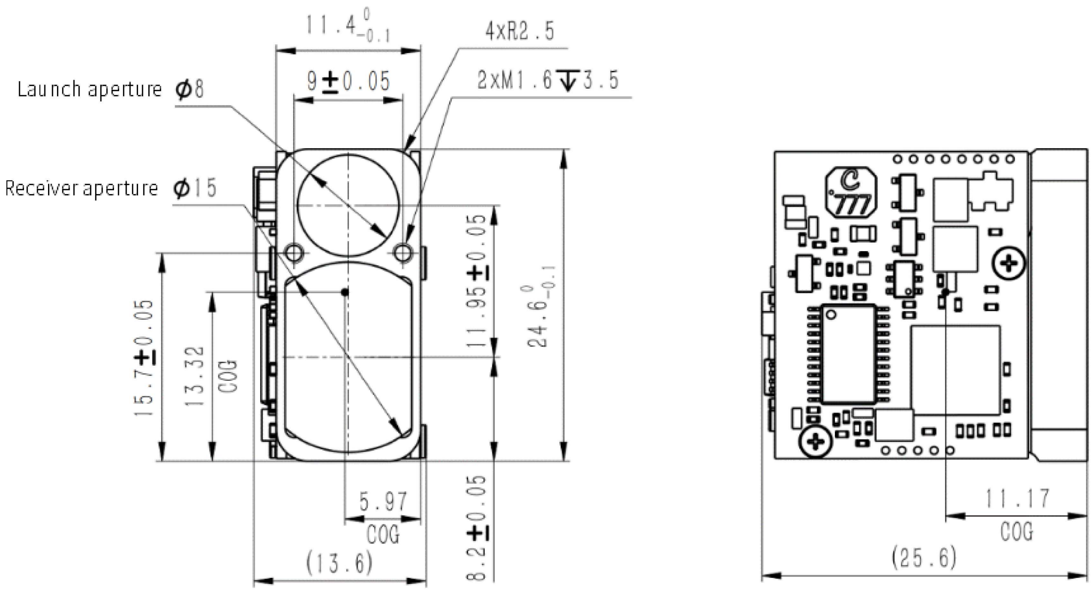
3.3 Cable Definition



Cable Specifications:

CN1（C-WB0812H-5-1-S-N）	CN2（C-WB0812H-6-1-S-N）	Cable Specifications	Color
1	6	AWG32	Black
2	5	AWG32	Red
3	3	AWG32	Yellow
4	2	AWG32	Green
5	1	AWG32	White

3.4 Mechanical Interface and Center of Gravity



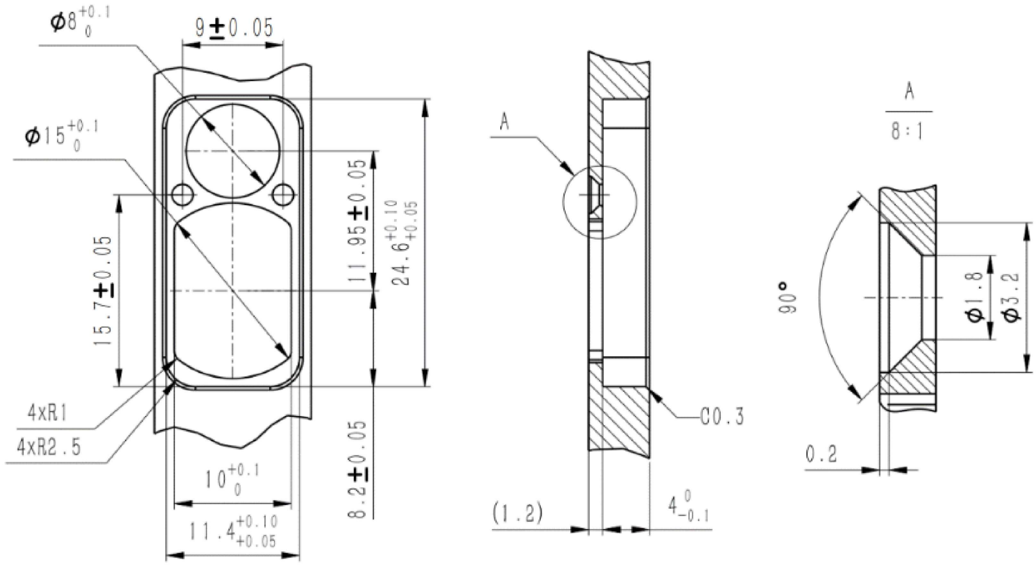
3.5 Precautions

3.5.1 Window Lens Coating

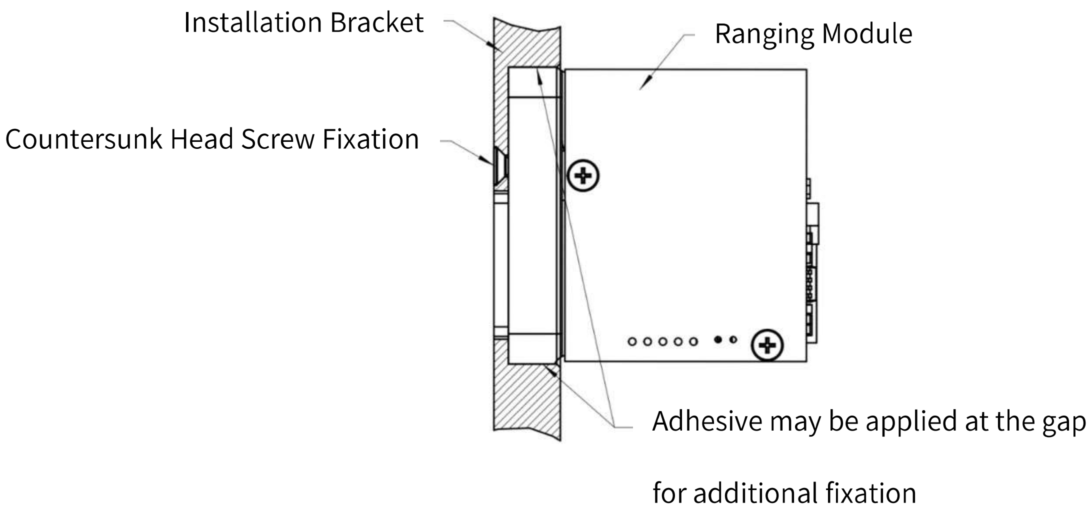
If window lenses are used for the integration of the entire machine, it is recommended that the transmittance of the window lenses for the 905 ± 20nm band be ≥ 98%.

3.5.2 Structural Installation

Recommended bracket dimensions:



Installation Diagram:



The thread length of countersunk screws shall not exceed 3.5mm.

4 Software Operation Instructions

4.1 Protocol Format

Baud Rate: 115200bps (factory default) / 57600bps / 38400bps / 9600bps

Data Format: 1 Start Bit, 8 Data Bits, 1 Stop Bit, No Parity

4.1.1 Data Packet Format

Field	Length	Range	Description
Frame Header	2	0xEE 0x16	Fixed Bytes
Data Length	1	2~7	Number of Subsequent Valid Bytes
Device Code	1	0x03	Fixed Bytes
Command Code	1	0x00~0xFF	Instruction Type Encoding
Related Parameters	0-5	0x00~0xFF	Parameter Value (command-dependent)
Check sum	1	0x00~0xFF	Sum of Device Code + Command Code + Parameters (low 8 bits)

4.2 Ranging Command

After powering on the distance measurement module, it defaults to a low POWER_ON state. It is necessary to pull POWER_ON high and wait for approximately 0.2 seconds (to allow the driving capacitor to charge) before proceeding with any subsequent command operations to avoid unresponsiveness from the distance measurement module. During the standby process, if POWER_ON is not pulled low, there is no need to wait for 0.2 seconds.

The following instructions are all hexadecimal.

4.2.1 Device Self-Diagnosis

Send: EE 16 02 03 01 04 (Transmit self-diagnosis command; return current working status information)

Response: EE 16 06 03 01 Status3 Status2 Status1 Status0 Check

To fulfill the requirements of bit parsing, convert hexadecimal to binary.

Status3：Reserved，Status2：Reserved

Status1：bit0--FPGA system status；　1 Normal　　0 Abnormal

bit5--Bias output status；　1 Bias normal　　0 Bias abnormal

bit6--Temperature status；　1 Temperature normal　　0 Temperature abnormal

Status0：Reserved

4.2.2 Single/Continuous Ranging

①Single Ranging

Send:EE 16 02 03 02 05

Response:EE 16 06 03 02 Status Result2 Result1 Result0 Check

②Continuous Ranging

Send:EE 16 02 03 04 07

Response:EE 16 06 03 04 Status Result2 Result1 Result0 Check

According to the bit parsing requirements, the hexadecimal needs to be converted to binary. Status: 0x00 indicates that the range measurement result is a single target; 0x04 indicates that the range measurement result exceeds the limit; Result2: high 8 bits of the range measurement value, Result1: low 8 bits of the range measurement value, Result0: decimal part of the range measurement value.

Distance Value = Result2 × 256 + Result1 + Result0 × 0.1, Unit: m

4.2.3 Stop Distance Measurement

Send:EE 16 02 03 05 08

Response:EE 16 02 03 05 08

4.3 Configuration Commands

4.3.1 Set Baud Rate

Send:EE 16 06 03 A0 BD24 BD16 BD8 BD0 Check

Response:EE 16 06 03 A0 BD24 BD16 BD8 BD0 Check

To parse by bit, convert hexadecimal to binary. The baud rate is of U32 type,with the high byte first: BD24 - BDO.

4.4 Other Quick Commands

4.4.1 Enter the spot brightness adjustment

Send:EE 16 04 03 10 01 power 校验 Check

Response:EE 16 04 03 10 01 power 校验 Check

Power represents the percentage of the configured laser brightness, with a data range of [1 to 100]. This command will affect the ranging frequency and human eye safety, and therefore should only be applied to the debugging process of spot inspection. When using normal distance measurement, this debugging mode cannot be enabled.

4.4.2 Exit the spot brightness adjustment (Enter the normal ranging mode)

Send:EE 16 04 03 10 00 00 13

Response:EE 16 04 03 10 00 00 13

4.4.3 Query FPGA Version

Send:EE 16 02 03 A6 A9

Response:EE 16 06 03 A6 Version1 Version2 Version3 Version4 校验 Check

4.4.4 Query MCU Version

Send:EE 16 02 03 A7 AA

Response:EE 16 06 03 A7 Version1 Version2 Version3 Version4 校验 Check

4.4.5 Query Hardware Version

Send:EE 16 02 03 A8 AB

Response:EE 16 06 03 A8 Version1 Version2 Version3 Version4 校验 Check

4.4.6 Query Device SN

Send:EE 16 02 03 A9 AC

Response:EE 16 06 03 A9 SN1 SN2 SN3 SN4 校验 Check

5 Routine Maintenance and Troubleshooting

Measured distance is shorter than expected	If the test is carried out on severe haze days, water mist days with heavy humidity,it is easy to detect water droplets or particles in the air, so it is recommended to choose a good test environment
Reduced ranging performance (e.g., failure to detect targets at theoretical maximum range)	①The extremely low reflectivity of the target will affect the return energy of the laser, which in turn will affect the ranging performance. When the ranging target is a low-reflection object such as a lake, a road surface in rainy weather, and glass, the actual performance of the product cannot reach the theoretical range ②The ranging module does not irradiate the target point vertically, but forms a certain angle, and the laser return energy in the oblique measurement state will attenuate, resulting in the reduction of ranging performance ③Due to the overlap between the laser wavelength and the solar spectrum, when the ranging module looks directly at the sunlight, it will lead to a decrease in ranging performance
Use our company's complete set of communication cables to communicate with the PC, and return without instructions	Check whether the communication cable is broken (the 6-pin communication cable is too thin and easy to break)
Use your own cable to	Check whether the external wiring sequence of the module is

communicate, no command to return	correct, whether the communication line is short-circuited, open or poorly contacted, and check whether the serial port TX and RX line sequences are reversed

6 Appendix

6.1 After-Sales Service Terms

1.Warranty Period: 12 months from the date of purchase.

2.Exclusions: Man-made damage, problems caused by improper use.